

# Algorithm parameters

The algorithm operates in three stages. First, geomagnetic storms are detected over the time period that the user has set. Second, the detected events are filtered, and their main phases are identified. Third, derived solar wind parameters are extracted for the filtered storms with data-complete main phases.

## Stage 1

The first stage identifies geomagnetic storms based on their Dst minima. At this stage, the program is concerned only with the minimum Dst depression of each storm and the time at which this minimum occurs.

**EPISODE\_LEVEL** defines the Dst threshold used to identify continuous disturbed intervals. A disturbed episode begins when Dst becomes equal to or lower than **EPISODE\_LEVEL** and continues as long as consecutive hourly Dst values remain at or below this threshold. Thus, **EPISODE\_LEVEL** acts as the initial, broad criterion for identifying magnetically disturbed periods. The default value is **−20 nT**.

**STORM\_LEVEL** defines the Dst threshold used to identify storm-level minima within each disturbed episode. After a disturbed episode has been detected, the algorithm searches within that episode for candidate storm minima. A **candidate storm minimum** is defined as a Dst value that is equal to or lower than **STORM\_LEVEL** and is also a local minimum within the surrounding time window defined by **LOCAL\_MIN\_RADIUS\_HOURS**. The default value is **−50 nT**, which is commonly used as the threshold for identifying moderate geomagnetic storms.

**LOCAL\_MIN\_RADIUS\_HOURS** defines the time window used to test whether a storm-level Dst value is a local minimum. For each Dst value that is equal to or lower than **STORM\_LEVEL**, the algorithm checks whether it is the lowest Dst value within a surrounding time window of  $\pm$ **LOCAL\_MIN\_RADIUS\_HOURS**. If it is, the point is retained as a candidate storm minimum. The default value is **3 hours**.

**MINIMA\_SPLIT\_FACTOR** defines the **minimum-splitting criterion**, which controls whether multiple candidate storm minima within the same disturbed episode are treated as separate storm events. When two candidate storm minima occur within the same episode, the algorithm examines the maximum Dst value between them, referred to here as the inter-minimum peak. The inter-minimum peak height is defined as

$$\text{peak height} = \text{inter-minimum peak} - \text{shallower minimum.} \quad (1)$$

This height is then compared with a separation threshold, defined as a fraction of the absolute depth of the deepest minimum:

$$\text{separation threshold} = -\text{deeper minimum} \times \text{MINIMA\_SPLIT\_FACTOR.} \quad (2)$$

If the peak height is greater than or equal to the separation threshold, the two minima are treated as separate storm events. Otherwise, they are considered part of the same storm structure, and the deeper minimum is retained as the representative minimum Dst of the storm. The default value for **MINIMA\_SPLIT\_FACTOR** is **0.4**.

If more than two candidate storm minima occur within the same episode, they are evaluated sequentially in chronological order. Each new candidate minimum is compared with the most

recently retained minimum. If the separation criterion is satisfied, the new minimum is retained as a separate storm event. If not, the two minima are merged and only the deeper minimum is retained before the algorithm proceeds to the next candidate minimum.

**STORM\_LIMIT** imposes a lower bound on accepted Dst minima. It allows the user to restrict the analysis to a selected storm-intensity range and to exclude events outside the desired limits.

**Note:** To select a specific storm-intensity range, **STORM\_LEVEL** can be used as the upper threshold and **STORM\_LIMIT** as the lower threshold. For example, setting **STORM\_LEVEL** =  $-50$  nT and **STORM\_LIMIT** =  $-99$  nT restricts the analysis to storms with  $-99 \text{ nT} \leq \text{Dst}_{\min} \leq -50 \text{ nT}$  (moderate storms).

## Stage 2

In the second stage, the program identifies the main phase of each storm detected in Stage 1. Because the end of the main phase is defined by the minimum Dst time,  $t_{\min}$ , Stage 2 focuses on estimating the corresponding onset time of the main phase,  $t_{\text{start}}$ .

For each detected storm, the algorithm searches backward in time from  $t_{\min}$ . This search uses a reference Dst value, referred to as the **mark**. Once the backward search reaches a Dst value equal to or higher than the mark, the first local Dst maximum encountered is assigned as  $\text{Dst}_{\text{start}}$ , and its timestamp is assigned as  $t_{\text{start}}$ . The main phase duration is then calculated as the time difference between  $t_{\text{start}}$  and  $t_{\min}$ .

The Stage 1 storm detection parameters are not reapplied during this stage. Stage 2 uses only the storm minima and their timestamps previously identified during storm detection. Consequently, the determination of  $t_{\text{start}}$  is not constrained by the temporal limits of the original disturbed episode. Instead, the algorithm searches backward through the available Dst record until both the mark condition and the local maximum criterion are satisfied. If no valid  $t_{\text{start}}$  can be identified, the storm is excluded from the filtered storm sample. In this study, filtered storms are defined as detected storms for which the main phase can be reliably identified.

**STRONG\_STORM\_THRESHOLD** separates strong storms from weaker storms for the purpose of defining the mark that will be used for main phase onset identification. The default value is  $-100$  nT. Storms with  $\text{Dst}_{\min}$  equal to or deeper than this threshold use the fixed mark value defined by **STRONG\_MARK**.

**STRONG\_MARK** is the mark used for strong storms. The default value is  $-50$  nT. During the backward search from  $t_{\min}$ , once Dst reaches or exceeds this value, the first local maximum encountered is assigned as  $\text{Dst}_{\text{start}}$ , and its timestamp is assigned as  $t_{\text{start}}$ .

For weaker storms, the mark is scaled according to storm intensity:

$$\mathbf{WEAK\_MARK} = \text{Dst}_{\min} \times \frac{\mathbf{STRONG\_MARK}}{\mathbf{STRONG\_STORM\_THRESHOLD}}. \quad (3)$$

With the default values, this gives  $\mathbf{WEAK\_MARK} = \text{Dst}_{\min} \times \frac{1}{2}$ .

**LOCAL\_MAX\_RADIUS\_HOURS** defines the time window used to identify a local Dst maximum during the backward search for the storm onset. A point is accepted as a local maximum if it has the highest Dst value within a surrounding time window of  $\pm \mathbf{LOCAL\_MAX\_RADIUS\_HOURS}$ . The default value is **3 hours**. Increasing this parameter makes the onset detection less sensitive to short-term Dst fluctuations. However, if the value is too large, an earlier local maximum may be

selected when pre-storm variations are present, resulting in an overestimation of the main phase duration.

After  $t_{\text{start}}$  has been estimated, the algorithm checks whether the Dst minimum of a previously detected storm occurs between  $t_{\text{start}}$  and  $t_{\text{min}}$  of the current storm. If such a previous minimum exists, the current storm is excluded from the filtered sample. This criterion prevents the algorithm from assigning a main phase interval that extends across an earlier storm minimum. In such cases, the estimated interval no longer represents the development of a single independent storm. This exclusion helps ensure that main phase duration and the subsequent analysis of relationships with solar wind parameters are calculated only for storms with a physically meaningful onset-to-minimum interval.

After the previous-storm exclusion has been applied, the program optionally applies a disturbance filter. This filter is applied primarily to low-intensity storm events in which the Dst evolution before  $\text{Dst}_{\text{min}}$  is irregular, containing multiple secondary dips. Such events are still geomagnetic storms, but they are separated from the filtered storm sample because their automatic  $t_{\text{start}}$  determination is less reliable. In an idealized storm evolution, Dst decreases relatively continuously from the main phase onset toward  $\text{Dst}_{\text{min}}$ . In practice, however, many storms show irregular pre-minimum Dst variations, including secondary dips before the final minimum. For low-intensity storms, these variations can make the automatic onset detection particularly ambiguous and may produce disproportionately long main phase durations, for example several tens of hours for a moderate storm with  $\text{Dst}_{\text{min}}$  near  $-60$  nT. The disturbance filter therefore separates these ambiguous events from the filtered storm sample.

**DISTURBANCE\_FILTER** activates or deactivates this filtering step.

**DISTURB\_LEVEL** defines the intensity range to which the disturbance filter is applied. Only storms with  $\text{Dst}_{\text{min}}$  shallower than **DISTURB\_LEVEL** are tested by the disturbance filter. With the default value of  $-100$  nT, the filter is applied to moderate storms.

**DISTURB\_DIP\_COUNT** defines the minimum number of secondary Dst dips required for a storm to be classified as a disturbance. If the number of dips between  $t_{\text{start}}$  and  $t_{\text{min}}$  reaches this value, the event is excluded from the filtered storm sample and added to the disturbance category. The default value is **2**.

**DISTURB\_DIP\_RADIUS\_HOURS** defines the time window used to identify secondary Dst dips during disturbance filtering. Between  $t_{\text{start}}$  and  $t_{\text{min}}$ , the algorithm checks whether each Dst value is the lowest value within a surrounding time window of  $\pm \text{DISTURB\_DIP\_RADIUS\_HOURS}$ . If it is, the point is counted as a secondary Dst dip. The default value is **3 hours**. A larger radius makes the filter less sensitive to short-term Dst fluctuations, reducing the number of detected secondary dips and making storms less likely to be classified as disturbances.

## Stage 3

In Stage 3, the program uses the OMNIWeb solar wind data within the main phase intervals identified after Stage 2 filtering to compute the corresponding solar wind derived parameters for each storm. Before computing the final parameters, the program checks the OMNIWeb solar wind data for missing values during the main phase of each storm. If a single value is missing for a given parameter, it is replaced by linear interpolation using the neighboring valid values. Storms containing consecutive missing values in any solar wind parameter during their main phase are rejected from the final data-complete sample. This step ensures that the derived solar wind

parameters are calculated only for storms with sufficiently complete input data.

For each data-complete storm, the derived solar wind parameters are computed over the identified main phase and include the IMF peak ( $\text{IMF}_p$ ),  $B_z$  peak ( $B_{zp}$ ), solar wind speed peak ( $V_p$ ),  $E_y$  peak ( $E_{yp}$ ) and integrated  $E_y$  ( $E_{yi}$ ), defined here as the sum of positive  $E_y$  values within this interval. The  $B_z$  peak is defined as the most negative  $B_z$  value, whereas the other peak quantities correspond to the maximum values reached.